

NEPHROLOGY TEACHING SERIES

Glomerulonephritis

Active sediment, pattern recognition, urgent biopsy

Hematuria + proteinuria + rising Cr is not just AKI — look at the urine.

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MS3/MS4 rotation teaching

Why we get consulted

Rapidly progressive GN can cause irreversible loss if not recognized early.

TRIGGERS FOR CONSULT

- Active sediment: RBC casts, dysmorphic RBCs
- Hematuria + proteinuria + rising Cr
- Nephrotic range (> 3.5 g/day or UPCR > 3.5 g/g)
- Pulmonary-renal syndrome
- Rash + arthritis + kidney disease
- Unexplained AKI with active UA
- Hypocomplementemia + AKI

TEACHING PEARL

Hematuria + proteinuria + rising Cr is never 'just AKI.' Urine findings can flip management from watchful ATN to urgent biopsy + immunosuppression.

Nephritic vs nephrotic — the orienting split

Two patterns, different differentials, different urgencies.

NEPHRITIC — inflammation

Hematuria (dysmorphic RBCs, RBC casts)

Modest proteinuria (usually < 3 g/day)

Hypertension (salt + water retention)

Rising Cr, often acute/subacute

Edema — periorbital, pedal

Think: IgA, post-infectious, lupus, antineutrophil cytoplasmic antibody (ANCA), anti-glomerular basement membrane (anti-GBM)

Urgency: rising Cr = consider rapidly progressive glomerulonephritis (RPGN) workup

NEPHROTIC — barrier loss

Proteinuria > 3.5 g/day (or UPCR > 3.5 g/g)

Hypoalbuminemia (< 3 g/dL)

Edema — often generalized, anasarca

Hyperlipidemia, hypercoagulability (renal vein thrombosis, DVT)

Bland sediment (± oval fat bodies, 'maltese cross')

Think: minimal change, focal segmental glomerulosclerosis (FSGS), membranous, DKD, amyloid

Urgency: usually less acute, but assess VTE risk, edema, and lipids

Proteinuria and nephrotic syndrome

Quantify first — then recognize the complications.

QUANTIFY PROTEINURIA

UACR — best screen (albumin specific)

Normal < 30 mg/g

Moderate A2 = 30–300

Severe A3 = > 300

UPCR (g/g) ≈ g/day proteinuria

Nephrotic range: > 3.5 g/day or UPCR > 3.5 g/g

24-h urine: gold standard but often impractical

NEPHROTIC SYNDROME — MANAGE

Tetrad: > 3.5 g/day proteinuria, albumin < 3, edema, hyperlipidemia

VTE risk — consider anticoagulation when albumin is very low, especially membranous, after bleeding-risk review

Infection risk (loss of Ig, complement)

Tx: max ACEi/ARB, SGLT2i, statin, Na restriction

Biopsy confirms primary cause — drives immunosuppression

UA sediment — reference card

Optional reference. Most labs report sediment findings automatically.

Finding	What it means
RBC casts / dysmorphic RBCs	Glomerular bleeding → GN (IgA, ANCA, lupus, anti-GBM, post-strep)
Muddy brown / granular casts	Tubular injury → ATN (ischemic, septic, toxic)
WBC casts	Pyelonephritis, acute interstitial nephritis (AIN)
Waxy casts	Chronic kidney disease (advanced, low urine flow)
Hyaline casts	Nonspecific — dehydration, fever; can be normal
Oval fat bodies / Maltese cross	Heavy proteinuria / nephrotic syndrome
Eosinophils	AIN (low sensitivity) or atheroembolic disease
Crystals (calcium oxalate, uric acid)	Ethylene glycol (oxalate); tumor lysis / urate; stones

Note: most outpatient labs report sediment automatically; manual microscopy is rarely done in clinic but remains the gold standard when the diagnosis is unclear.

The GN serologic panel

Send it early — results often take days, and timing matters.

Test	What it tells you	Disease
ANA, anti-dsDNA, anti-Sm	Lupus activity; dsDNA correlates with nephritis	SLE nephritis
C3 / C4	Low in SLE, post-strep, membranoproliferative glomerulonephritis (MPGN), cryoglobulinemia; normal in ANCA and anti-GBM	Narrow ddx quickly
ANCA (myeloperoxidase (MPO), proteinase 3 (PR3))	PR3 → granulomatosis with polyangiitis (GPA); MPO → microscopic polyangiitis (MPA); either can cause renal-limited vasculitis	ANCA-associated vasculitis
Anti-GBM	Linear IgG on biopsy; rapid and irreversible	Goodpasture syndrome
PLA2R antibody	Positive in ~70% of primary membranous; follow to monitor therapy	Membranous nephropathy
HBV, HCV, HIV, cryoglobulins	Secondary causes of MPGN or membranous	Infection-driven GN
Complement CH50	Screen for C3 glomerulopathy / dense deposit disease	Rare complement-mediated GN
ASO, anti-DNase B	Recent strep infection (1–3 wk throat, 3–6 wk skin)	Post-streptococcal GN

Urgent: pulmonary-renal syndrome & RPGN

Minutes-to-hours decisions. Biopsy and immunosuppression don't wait.

RED FLAGS

Hemoptysis + hematuria + rising Cr
New oxygen requirement, infiltrates on CXR
Cr doubling over days; oliguria
Active sediment (RBC casts)
Mononeuritis multiplex, sinus disease, upper-airway bleeding
ANCA or anti-GBM positive
Action: biopsy TODAY if feasible; empiric steroids + consider PLEX

INITIAL MANAGEMENT (before biopsy)

Stabilize: O2, transfuse if bleeding, protect access
Send: ANCA, anti-GBM, ANA, dsDNA, C3/C4, urgent UA + microscopy
Empiric IV methylprednisolone 500–1000 mg/day × 3 if anti-GBM suspected (can't wait)
PLEX: consider for severe kidney disease, rapidly rising Cr, dialysis need, or hypoxemic pulmonary hemorrhage
Biopsy as soon as platelets, BP, anticoagulation allow
[Consult IR / nephrology for urgent native kidney biopsy](#)

Common glomerular diseases — recognition

Pattern + clinical context = diagnosis most of the time.

Disease	Presentation	Clue
IgA nephropathy	Recurrent synpharyngitic hematuria in young adult	IgA-dominant IF on biopsy; most common GN worldwide
Post-infectious GN	Kids 1–3 wk after strep throat/impetigo; adults post-staph	Low C3 (recovers 6–8 wk); humps on EM
Lupus nephritis	Young woman, rash, arthritis, serositis, cytopenias	ANA+, dsDNA+, low C3/C4; class I–VI on biopsy
ANCA vasculitis (GPA, MPA)	Older adult, sinus/pulmonary disease, RPGN	MPO/PR3+; pauci-immune crescentic GN
Anti-GBM (Goodpasture)	Hemoptysis + RPGN, bimodal age	Linear IgG; urgent PLEX + immunosuppression
Membranous nephropathy	Adult nephrotic syndrome + VTE	PLA2R+ in ~70% (primary); spike & dome on EM
FSGS	African ancestry/APOL1 risk, obesity, HIV, heroin, reflux — nephrotic range proteinuria	Segmental sclerosis; APOL1 variants
Minimal change	Sudden full nephrotic syndrome; children; NSAID-associated in adults	Normal LM, foot-process effacement on EM

Kidney biopsy + treatment scaffolding

Biopsy confirms; supportive care + targeted immunosuppression treats.

WHEN TO BIOPSY

Rapidly rising Cr + active sediment

Nephrotic syndrome in adult (most — exclude DKD)

Unexplained AKI with hematuria/proteinuria

Suspected lupus nephritis to classify

Isolated persistent proteinuria > 1 g/day with features

Pre-biopsy: platelets > 50K, INR < 1.5, BP < 160/90, stop AC 5–7 d

Complications: bleeding (~2%), transfusion (~1%), AV fistula, rarely nephrectomy

SUPPORTIVE THERAPY (always)

Max-dose ACEi or ARB for proteinuria

SGLT2i (added benefit in proteinuric CKD — DAPA-CKD, EMPA-KIDNEY)

BP target: standardized office SBP < 120 when tolerated; routine/home targets individualized

Low-sodium diet (< 2 g/day)

Statin for lipid management (nephrotic)

Consider anticoagulation for high-risk nephrotic syndrome after bleeding-risk review

Vaccinations before immunosuppression (PPSV, HBV, influenza, COVID, zoster)

Landmark trials in glomerular disease

The RCTs that shape current management.

Trial	Year	Takeaway
RAVE	2010	Rituximab non-inferior to cyclophosphamide for ANCA induction; preferred in relapse.
MAINRITSAN	2014	Rituximab maintenance superior to azathioprine in ANCA vasculitis.
PEXIVAS	2020	Reduced-dose steroids non-inferior; PLEX not routine, but KDIGO 2024 still considers it in highest-risk AAV.
MENTOR	2019	Rituximab SUPERIOR to cyclosporine at 24 mo for membranous (durable remission).
STOP-IgAN	2015	Adding immunosuppression to supportive care did NOT slow IgAN; KDIGO 2025 now pairs SC with IgA-targeted therapy.
TESTING	2022	Methylpred slowed IgAN progression (HR 0.53 full cohort; HR 0.27 reduced-dose); more infection.
NeflgArd	2023	Targeted-release budesonide ↓ proteinuria & slowed eGFR decline in IgAN.
DAPA-CKD	2020	Dapagliflozin ↓ kidney failure even in non-DM CKD (including IgAN, FSGS subgroups).
EMPA-KIDNEY	2023	Empagliflozin benefit across eGFR/urine albumin-to-creatinine ratio (UACR), including many primary GN etiologies.

Common presentation mistakes

Avoid these when presenting a possible GN.

AVOID

- Calling it AKI without checking UA
- Relying on dipstick alone
- Missing day-1 serologic panel
- Delaying biopsy for labs
- Steroids without a taper plan
- Automatic anticoagulation for every nephrotic patient
- No vaccines before immunosuppression

DO INSTEAD

- Lead with UA + proteinuria
- Request formal UA + microscopy
- Day 1: ANA, dsDNA, ANCA, anti-GBM, C3/C4
- Tee up biopsy early (plts, INR, BP)
- Pair steroids with rituximab/CYC + taper
- Low albumin → weigh VTE vs bleeding risk
- Vaccinate before steroids

Clinical case

Think it through — answer is on the next slide.

VIGNETTE

A 48-year-old man, 3-week history of sinusitis, hemoptysis, and fatigue. Admitted with Cr 3.4 (baseline 1.0 six months ago), UA: 3+ blood, 2+ protein, RBC casts, dysmorphic RBCs; UACR 2800. BP 168/102. CXR: bilateral patchy infiltrates. C3 and C4 normal. ANCA pending. No rash.

QUESTION: What's the likely diagnosis, what do you send today, and what's the urgency?

Clinical case — answer

Key teaching points from the case.

ANSWER

ANCA-associated vasculitis (likely GPA: sinus + pulmonary + renal, normal complements). Send ANCA (MPO/PR3), anti-GBM, ANA/dsDNA. Biopsy TODAY. Empiric high-dose IV methylprednisolone after samples drawn.

TEACHING

Pulmonary-renal syndrome with rapidly rising Cr + active sediment = RPGN until proven otherwise. Classic triad for GPA: upper airway (sinus, nose) + lower airway (lung) + kidney. Normal C3/C4 argues against lupus or post-infectious. While waiting for biopsy: stabilize BP, oxygenation, no anticoagulation if biopsy planned. RAVE showed rituximab non-inferior to cyclophosphamide; PEXIVAS supports reduced-dose steroids. PLEX is not routine, but KDIGO 2024 still considers it for SCr > 3.4 mg/dL, dialysis/rapidly rising SCr, hypoxemic alveolar hemorrhage, or ANCA/anti-GBM overlap.

REFERENCES

Glomerulonephritis

Guidelines

KDIGO 2021 Glomerular Diseases Guideline
KDIGO 2024 ANCA Vasculitis Guideline
ACR 2019 Lupus Nephritis Guidelines
EULAR/ERA-EDTA SLE Management Recs
CHAMP — ANCA vasculitis management consensus
ASN/NephJC educational resources

Landmark Trials

RAVE — NEJM 2010
MAINRITSAN — NEJM 2014
PEXIVAS — NEJM 2020
MENTOR — NEJM 2019
STOP-IgAN — NEJM 2015
TESTING — JAMA 2022
NeflgArd — Lancet 2023
DAPA-CKD — NEJM 2020
EMPA-KIDNEY — NEJM 2023

UpToDate (2026 access):

Overview of the classification and causes of glomerular disease • Evaluation of the adult with hematuria • Approach to the patient with pulmonary-renal syndrome • Treatment of ANCA-associated vasculitis • Treatment of lupus nephritis • Primary membranous nephropathy: pathogenesis and treatment